

Paper 3 — The Comprehension Section

Mathematics B (MEI) — OCR B — A-Level (H640) — Paper 3 (H640/03), Section B — how to revise for it

What it is. H640/03 is *Pure Mathematics and Comprehension*, 75 marks in 2 hours. Section A is 60 marks of ordinary pure maths. **Section B is 15 marks** and is based on an **article printed on a separate Insert**, which you get with the paper. You read the article, then answer questions on it. The questions refer to the article by **line number**.

This is the one part of the A-Level that isn't a topic, so it is easy to leave unrevised. It can be revised, and this page is how.

1. The maths is ordinary A-Level content

This is the single most important thing to understand. The article supplies the *context* — a piece of mathematics you have not met before. It does not supply new techniques. Every question is answered with content you have already been taught, and the article is written on the assumption that you know that content cold.

Here is what the eight published articles were about, and the A-Level content each one actually required:

Series	The article was about...	...but the maths you needed was
June 2018	The inequality of arithmetic and geometric means	Proof from $(a - b)^2 \geq 0$, geometric sequences, angle in a semicircle, trigonometry in similar triangles, proof by contradiction
June 2019	The volume of a toothpaste tube	Equation of a straight line, algebraic fractions, integration as the limit of a sum, testing a model against a real measurement

Nov 2020	When is a^b bigger than b^a ?	Indices and logarithms, $\int \frac{1}{x} dx$, increasing functions, quotient rule, stationary points and the second derivative
Nov 2021	Writing π as a sum of arctangents	The $\tan(A + B)$ formula, the graph of $\arctan x$, algebraic manipulation of fractions
June 2022	Approximations to the sine function	Differentiation, stationary points, change of sign, substitution, compound angle formulae
June 2023	Euler's approximate formula for a sum	Integration, sigma notation, sum of a series, logarithms, percentage error
June 2024	Where tangents and normals to a quadratic curve intersect	Differentiation, equations of tangents and normals, coordinate geometry, algebraic manipulation
June 2025	The curve $y^2 = \frac{x^2(3a - x)}{a + x}$	Implicit and parametric curves, $\tan 3\theta$ from compound angles, inequalities, asymptotes

What this means for revision. There is no separate body of knowledge to learn for Section B. Strong, secure knowledge of the Pure content is the preparation. If differentiation, logs, compound angles, series and proof are solid, the comprehension is a reading exercise. If they are shaky, no amount of reading practice will rescue it.

2. Most of Section B is "show that"

Across the eight published papers, Section B contained 57 questions. Their opening words are strikingly repetitive:

- **19** begin "Show that..."
- **5** begin "Hence...", **4** "Use...", **3** "Prove...", **3** "Verify..."
- **60%** quote a specific line number from the article

Counting every *show*, *prove* and *verify* question together: **63% of Section B parts, and 72% of Section B marks, give you the answer in the question.**

The consequence. In a "show that" question the answer is printed, so **none** of the marks are for the answer. Every mark is for the working that gets there. Mark schemes for these parts say "*Convincing completion*", "*at least one step of working must be shown*", "*no incorrect work seen*" and "*must not work backwards*".

Three ways students lose these marks.

- **Working backwards** — starting from the printed answer and arriving at something true. This scores zero on most of these parts. Start from what you are given and finish at the printed result.
- **Jumping steps** — going from the first line to the answer because "it's obvious". The intermediate line *is* the mark.
- **Deciding it looks too hard** — because the context is unfamiliar. Look at what is being asked, not at the article's subject matter.

3. Working with the Insert

The article is a separate booklet. You are told to read it before attempting the questions, and Section B is written assuming you have.

- **The line numbers matter.** Six questions in ten quote one. "...as given in line 24" means the result is already printed in the article — you are being asked to justify it, not discover it. Find the line, and you will know exactly what you are aiming at.
- **Results in the article are tools you are expected to reuse.** A later question will hand you an earlier result to work with. In June 2025, a question quoted a formula from line 23 and asked you to use it; in June 2018, the final question said "Using this fact..." about a statement in lines 31–32. You are not expected to prove those from scratch — use them.
- **The article defines its own notation.** Letters such as a , b , h , r , t and θ mean whatever the article says they mean. Check the definition rather than assuming.
- **Read it once for the shape of the argument, then again for detail.** You do not need to follow every line on the first pass. You need to know what the article is trying to establish and roughly how, so that when a question points at line 24 you know where you are.

4. The question types that keep coming back

Type	Looks like	What to do
Justify a printed result	"Show that the equation of the line in Fig. C2 is $ry + hx = hr$, as given in line 24."	Work forwards to the printed result. Show the intermediate step.
Check a formula in a special case	"Verify that the formula is also valid for Fig. C3.1."	Substitute the special value, evaluate, and say why the result is right.
Reuse an article result	"Use the arctan addition formula in line 23 to show that..."	Quote the formula from the article, then substitute. Do not re-derive it.
Apply it to numbers	"A typical tube measures 5.4 cm across and is 12 cm high... comment on the accuracy of the formula."	Do the arithmetic, then <i>compare</i> and comment. The comment carries a mark.
Probe a limitation	"Explain why the result cannot be extended to apply when both a and b are negative."	Find the step in the argument that breaks. Usually a square root, a division, or a sign.
Prove it properly	"Prove by contradiction that..."	Full proof structure: assume the opposite, derive the contradiction, state the conclusion.

5. Practise reading mathematics

The skill Section B tests, beyond the content, is **reading unfamiliar mathematics at speed and doing something with it**. That is trainable, and it is trained by reading mathematics you have not seen before — not by doing more exercises on topics you already know.

- **The sample and past Inserts** are the closest thing to the real exam. Work through them properly: read the article, answer the questions, then mark yourself against the scheme paying attention to how much working the "show that" parts demanded.
- **Plus Magazine** (plus.maths.org) publishes exactly this kind of article — a short piece of real mathematics written for a reader who has not met it before. That is the Insert's register almost exactly.

- **NRICH** (nrich.maths.org) is good for the other half of the skill: being handed an unfamiliar problem and having to decide which technique applies.

Read actively. Reading an article through and understanding it is not the same as being able to use it under time pressure. When you read one, stop at each result and ask: *where did that come from, and could I justify it if a question told me to?* That is the exact question Section B asks.

6. A revision routine that works

1. **Get the Pure content secure first.** It is 60 of the 75 marks on this paper anyway, and it is what Section B is built on.
2. **Work every past Insert you can get** — there are eight published series plus the sample. Do them timed: about 30 minutes for reading and Section B together.
3. **Mark yourself strictly on the "show that" parts.** Ask whether an examiner would call your working a convincing completion, not whether you got the right answer — you were given the right answer.
4. **Read one Plus or NRICH article a week** between papers, to keep the reading muscle working.
5. **Keep a list of the article results you were asked to reuse.** You will start to recognise the pattern: the article always builds one or two key results, and the questions always send you back to them.

In one line. Section B is not a comprehension test with some maths in it. It is a maths test with some comprehension in it — and nearly three quarters of its marks are awarded for working towards an answer you have already been given.